

The length invariant X_n of a folded sign field

How pairing reduces MED-CAP to an edge-return, and the precise obstruction that remains

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Abstract

The median-cap inequality $\text{med}_i \leq \frac{8}{3} \text{sep}_i$ on a tiled block packing is equivalent to a constraint X_n on the length sequence (n_i) . Interval geometry does not prove X_n : the tiled composition $n = (1, 2, 8^{10})$ violates it. This note proves the combinatorial half. Blocks are paired consecutive same-sign runs of a $\pm$ colouring of a tiled integer interval, with an odd tail its own block, pairing from x -min. Then $n = 1$ occurs only as that odd tail; the tail sits at θ -min; the prefixes $(2, 1)$ and $(1, 1)$ are impossible; and the unique possible locus with $\text{sep}_i = \frac{3}{2}$ is the edge pair $(n_0, n_1) = (1, 2)$. Interior small-separation does not exist. Pairing does *not* forbid C_2 : random paired run sequences still violate MED-CAP in a few percent of draws. What pairing does is isolate the only pattern that can violate — a length-1 tail, two further singleton runs, and a subsequent plateau of long runs. On the source construction that pattern occurs once among 134 windows (15 428 atoms): the saturating prefix $(1, 2, 6, 5, 4, 4)$ of χ_4 at index 53, with the nearest $n \geq 7$ at index 7 (outside the radius-5 window). The named remainder is a local spacing statement for the bordered-chain polynomial v_2 at the 20% x -mask: after three consecutive singleton runs the next four runs must not form a plateau of length ≥ 3 . The minimal grid-consistent violator is the run sequence ending $(\dots, \geq 3, \geq 3, 1, 1, 1)$. It is not realised on the surface. Nothing in this document is evidence for or against the Riemann hypothesis.

Status. Lemmas 1–4 are proved (construction algebra and pairing combinatorics). Lemma 5 is proved (pairing does not kill C_2). Lemma 6 is proved (the order-statistic content of X_n at the only possible small-sep locus). Proposition 1 is a finite census, not a cofinal law. The remainder is named in §6. On the 134-window surface, X_n holds, hence MED-CAP holds, hence the pointwise floor $g_i \geq 3/8$ holds, hence T'_2 trivialises. The T1 constant C_K is not touched. No RH claim.

1 Recollection

We use the packing notation of [1] without change. Blocks $I_i \subset \mathbb{Z}$ of lengths n_i are pairwise disjoint and ordered; c_i is the unweighted centre; sep_i is the cut-geometry lower bound $\min(n_i + n_j)/2$ on adjacent sides; med_i is the radius- $W = 5$ window median of neighbouring gaps. MED-CAP is $\text{med}_i \leq \frac{8}{3} \text{sep}_i$ for every i . On a *tiled* packing (no unoccupied integer between or inside blocks) one has $\text{gap}_i = \text{sep}_i$ identically, and MED-CAP becomes a statement about (n_i) alone. That tiling identity holds on 134 windows / 15 428 atoms of the source [1]. The obstruction X_n is that (n_i) must not realise, at a small-separation locus ($\text{sep}_i = \frac{3}{2}$ from an adjacent $(1, 2)$ or $(2, 1)$, or $\text{sep}_i = 1$ from two adjacent singletons), a radius-5 window whose third-smallest neighbour-gap (edge) or mean of the 5th and 6th (interior) exceeds $\frac{8}{3} \text{sep}_i$.

The blocks of the source are not an abstract composition. They are read off a $\pm$ colouring of the tiled θ -grid, as follows.

2 The sign field

A window of depth N carries two discrete measures on $[-1, 1]$: the arithmetic comb (prime-power, Dirichlet, or second-family) and a smooth border comb. Each is obtained by the fold

$$j \mapsto \min(j, L - j), \quad x = \cos(2\pi \min(j, L - j)/L),$$

applied separately to the positive and negative cells of a lag-density d_{arm} on a circular grid of $L = 2M - 2$ points (μ keeps $d_{\text{arm}} > 0$, ν keeps $d_{\text{arm}} < 0$). Fold multiplicity is at most 2 (the two preimages j and $L - j$). The mesh identity of the frame ($h - \text{NU } u \in (0, \frac{3}{2}]$) makes the θ -grid $t = \arccos(x) \cdot (4N - 2)/(2\pi)$ integral on every surviving atom.

Level-2 blocks are defined on the *smooth border* bulk, after a 20% x -hull edge mask. Let c_t be the bulk contributions $c_t = w x v_2 \phi$, where w is the folded smooth weight ($w > 0$ on μ , $w < 0$ on ν), x is the fold coordinate, v_2 is the degree- $(N - 2)$ bordered-chain polynomial evaluated on the border nodes, and ϕ is a global scalar. Maximal same-sign runs of c_t in increasing- x order are paired from offset 0; an odd tail is its own block. Window (arithmetic) atoms are then assigned to those blocks by x -breakpoint search. The length n_i is the number of distinct θ -grid integers in block i .

Lemma 1 (sign identity and occupation). *On the four $w = 9$ worlds and on χ_4 at index 53,*

1. $\text{sign}(c_t) = \text{sign}(w) \text{sign}(x) \text{sign}(v_2)$ at every bulk atom (0 disagreements among $150 + 74 + 150 + 150 + 496$ atoms);
2. n_i equals the number of distinct θ -grid integers occupied by smooth-border atoms of the block — arithmetic window atoms add zero cells;
3. n_i equals the sum of the paired run lengths of that colouring, measured in distinct grid integers.

Consequently the length sequence is a function of the $\pm$ colouring of the tiled θ -grid of the smooth border, and of nothing else.

The identity in (i) is the product rule for $c_t = w x v_2 \phi$. Parts (ii)–(iii) are the fold-group ledger against the pairing definition; they are re-gated by exact integer counting in the companion script. On the $w = 9$ worlds the smooth weight w has a single sign-run on the whole bulk, so the colouring is $\text{sign}(v_2)$ times a global sign times $\text{sign}(x)$ (one flip at $x = 0$). At χ_4 index 53 the weight w has fourteen sign-runs; v_2 still dominates (run-length mode 2).

3 What pairing proves

Fix a finite sequence of positive integers $(r_0, \dots, r_{\nu-1})$, the grid lengths of consecutive same-sign runs in increasing- x order. Pairing from offset 0 produces the x -order lengths

$$n_j^x = r_{2j} + r_{2j+1} \quad (2j + 1 < \nu), \quad n_{\text{last}}^x = r_{\nu-1} \text{ if } \nu \text{ is odd.}$$

Because $x = \cos \theta$ is decreasing in θ on $[0, \pi]$, x -max is θ -min. The odd tail is therefore block 0 in θ -order, and the θ -order sequence is the reverse of the x -order sequence.

Lemma 2 ($n = 1$ only as odd tail). *If every $r_k \geq 1$, then $n_i = 1$ occurs if and only if ν is odd, $r_{\nu-1} = 1$, and $i = 0$ in θ -order. In particular $n = 1$ cannot occur in the interior, and cannot occur as a paired block.*

Proof. A paired block is a sum of two positive integers, hence at least 2. The only remaining possibility is the odd tail, of length $r_{\nu-1}$. That tail is θ -index 0. $\square$

Lemma 3 (forbidden prefixes). *The θ -order prefixes $(n_0, n_1) = (2, 1)$ and $(1, 1)$ cannot occur.*

Proof. $(1, 1)$ would require a tail of length 1 and a previous pair summing to 1. Two positive integers cannot sum to 1. $(2, 1)$ would require a tail of length 1 at θ -index 1, contradicting Lemma 2. $\square$

Lemma 4 (unique small-separation locus). *On any tiled packing obtained by pairing a positive run sequence, $\text{sep}_i \leq \frac{3}{2}$ occurs if and only if $(n_0, n_1) = (1, 2)$, and then only at $i = 0$ and $i = 1$, with $\text{sep}_0 = \text{sep}_1 = \frac{3}{2}$. In particular $\text{sep}_i = 1$ (two adjacent singletons) is impossible.*

Proof. $\text{sep}_i = \frac{1}{2}(n_i + n_j)$ for an adjacent pair, so $\text{sep} \leq \frac{3}{2}$ forces $\{n_i, n_j\} \in \{\{1, 1\}, \{1, 2\}\}$. Lemma 2 puts every 1 at index 0; Lemma 3 excludes $(1, 1)$ and $(2, 1)$. The remaining possibility is $(n_0, n_1) = (1, 2)$. $\square$

Thus X_n — which constrains the radius-5 window at small-separation loci — is a statement about a single edge, when that edge exists, and is vacuous on every packing whose odd tail is longer than 1 or whose run count is even.

Lemma 5 (pairing does not forbid C_2). *The run sequence whose θ -order lengths are $n = (1, 2, 8^{10})$ is a legal pairing of positive run lengths (tail 1, previous pair $1 + 1$, then pairs summing to 8). The tiled packing of that sequence violates MED-CAP at $i = 0, 1$ with ratio $16/3$. Among 1259 random paired sequences of run lengths in $\{1, \dots, 8\}$, a positive fraction (here 32) violate MED-CAP. A Chebyshev colouring $\text{sign}(\sin m\theta)$ on a uniform θ -grid, after the same 20% x -mask, produces 0 violations among 123 windows and 0 prefixes $(1, 2)$.*

The Chebyshev comparison is the reason the bulk mode is $n = 4$: zeros uniform in θ with spacing π/N against a grid of spacing $\pi/(2N)$ give run length 2, and pairing two such runs gives 4. An x -mask of a Chebyshev colouring cuts one run into a stub and leaves prefix $(1, 4, 4, \dots)$ typically, never $(1, 2)$. The prefix $(1, 2)$ is a *spacing gradient* at the mask — three consecutive singleton runs — which a uniform-in- θ colouring does not produce.

4 What X_n says at the only possible locus

Lemma 6 (order-statistic content of X_n). *Let $n = (1, 2, n_2, \dots)$ be tiled, $m \geq 6$. Write g_i for the gaps (equal to sides). Then $g_0 = g_1 = \frac{3}{2}$, and the five neighbour-gaps of $i = 0$ are $(g_1, g_2, g_3, g_4, g_5)$. MED-CAP at $i = 0$ is equivalent to the third-smallest of those five being ≤ 4 . In particular:*

- the saturating prefix $(1, 2, 6, 5, 4, 4)$ has third-smallest 4 and ratio $8/3$;
- a single spike $(1, 2, 7, 4, 4, 4)$ likewise has third-smallest 4 (an isolated $n = 7$ with immediate return to the bulk mode does not violate);
- the short C_2 prefix $(1, 2, 6, 6, 6, 6)$ has third-smallest 6 and ratio 4;
- the long C_2 prefix $(1, 2, 8^{10})$ has third-smallest 8 and ratio $16/3$.

Hence X_n at its only possible locus forbids a plateau of long runs after the $(1, 2)$, not a single overshoot.

Proof. Direct computation of sides, gaps, and the third-of-five, as in [1] Lemmas 3–4, plus the spike $(1, 2, 7, 4, 4, 4)$: sides $\frac{3}{2}, \frac{9}{2}, \frac{11}{2}, 4, 4$; neighbour-gaps of $i = 0$ sort as $(\frac{3}{2}, 4, 4, 4, \frac{9}{2})$; the third is 4. $\square$

The medcap formulation “ $n \geq 7$ at that locus is forbidden” is therefore slightly too strong as a *sufficient* picture: a lone 7 with return to 4 saturates. It is the right picture *in substance*, because a 7 that does not return produces three neighbour-gaps > 4 and violates. The precise statement is Lemma 6.

5 The source surface

Proposition 1 (one-shot census). *On the 134 windows of [1] Proposition 5 (42-rung prime-power core + 8 second-family anchors + 42 + 42 Dirichlet matchings; 15 428 blocks; tiling exact, 0 empty sides, 0 hull holes, 0 MED-CAP violations):*

1. $n = 1$ occurs 32 times, always at θ -index 0, always as an odd tail of length 1 (Lemma 2 with no exceptions);
2. of those 32, the next length is $n_1 = 4$ in 24 windows (the Chebyshev stub), $n_1 = 5$ in four, $n_1 = 8$ in two, $n_1 = 3$ in one, and $n_1 = 2$ in one;
3. that one $(1, 2)$ is χ_4 at index 53, prefix $(1, 2, 6, 5, 4, 4, 4, 8)$, third-smallest neighbour-gap 4, ratio $8/3$, nearest $n \geq 7$ at index 7 (distance 7 from $i = 0$, outside the radius-5 window);
4. consequently there are exactly two small-separation atoms on the whole surface (indices 0 and 1 of that window), and X_n holds at both, saturating.

The block-length histogram is $n = 4$ on 9006 atoms (58.4%), with a right tail out to $n = 16$ (once). Long blocks exist, in the interior. They are never adjacent to a $(1, 2)$.

The 114 atoms of [1] Proposition 6 — blocks with $n_i \leq 2$ that see some $n_j \geq 7$ inside the radius-5 window — are almost all interior $n = 2$ against a generous separation $\text{sep} \geq \frac{5}{2}$. They are not small-separation loci, and Lemma 4 says they cannot be. The worst such ratio $12/5$ is at χ_3 index 20, prefix $(1, 4, 8, 4, 8, 4)$, $\text{sep} = \frac{5}{2}$.

6 The remainder

Lemmas 2–4 reduce X_n to a single edge-return. Proposition 1 verifies the return on every window the construction currently produces. What is *not* proved as a cofinal law is that a deeper (or hypothetical) window cannot realise the plateau.

Minimal grid-consistent violator. The run sequence in x -order ending

$$(\dots, r, r, 1, 1, 1) \quad \text{with } r \geq 3$$

pairs to θ -order $n = (1, 2, 2r, \dots)$. Already $r = 3$ gives the short C_2 prefix $(1, 2, 6, 6, \dots)$ once the plateau continues for four runs; $r = 4$ gives $(1, 2, 8, 8, \dots)$. This colouring is legal as a $\pm$ assignment of a tiled interval, legal as a pairing, and legal as the restriction of a degree- $(N - 2)$ polynomial (a colouring with k runs needs $k - 1$ zeros). Fold multiplicity ≤ 2 and the mesh identity do not forbid it: they constrain how many source atoms land on a cell, not the run lengths of v_2 .

Named condition V_2 . Let v_2 be the degree- $(N - 2)$ bordered-chain polynomial of the window. Write R_{mask} for the run lengths of $\text{sign}(v_2 w x)$ on the bulk θ -grid, in x -order. Condition V_2 is: if the last three values of R_{mask} are $(1, 1, 1)$, then among the four preceding run lengths at least two are ≤ 2 (no 6-plateau at the mask). Equivalently, v_2 has no local $3 \times$ dilation of zero spacing immediately after a triple sign-flip at the 20% x -mask.

V_2 is the OP-spacing analogue (Freud / arcsine equilibrium spacing of consecutive zeros) for this specific polynomial. It is *not* a cited classical theorem: v_2 is a bordered-chain polynomial, not the orthogonal polynomial of a fixed positive continuous measure. A Chebyshev colouring satisfies V_2 vacuously (it never produces three consecutive 1-runs after an x -mask). The source v_2 satisfies V_2 on all 134 windows. The bulk mode $n = 4$ is the global counterpart of the same spacing (average run length 2, pairing to 4).

Proposition 2 (conditional MED-CAP). *On every tiled packing produced by the construction of §2,*

$$X_{\text{tile}} + \text{Lemmas } 2\text{--}4 + V_2 \implies X_n \implies \text{MED-CAP}.$$

Hence, on that packing, $g_i = \text{gap}_i / \text{med}_i \geq 3/8$ pointwise, $S_r = \sum_{g_i \leq 2^{-r}} q_i = 0$ for every $r \geq 2$, and T'_2 is trivial. The cubic-mass bound $M_3 \leq (\frac{8}{3})^2 C_K^2 (\log m)^2 / m^2$ of [1] §T1 carries, still carrying C_K .

On the 134-window surface V_2 is verified, so the implications hold unconditionally *there*. A cofinal theorem would need V_2 as a lemma about the bordered chain, or a replacement that bounds consecutive v_2 -zero spacings at the mask. Degree counting alone does not give V_2 : a degree- $(N - 2)$ polynomial may cluster its zeros and leave a gap.

7 T1

The conclusion of [1] §T1 stands. X_n is a constraint on run lengths of a sign field. The masses q_i see the same field as an amplitude. Proving X_n , even cofinally, does not produce a family-uniform C_K . The r358 record (per-family freeze maximum fails on the test set; ceiling 23.70 against freeze 4.91) is unchanged. After MED-CAP, T1 remains the single quantitative rest of the terminal lane.

8 Machine checks

Every numbered lemma of §§3–4 is re-proved by exact arithmetic in the companion script `verify_xn_steps.py` (same directory), together with the sign identity, the occupation identity, and the χ_4 index-53 pin of Proposition 1 (iii). The 134-window census of Proposition 1 is a one-shot enumeration against the same Fractions ledger (not a sealed probe), recorded here. Exit line: XN STEPS VERIFIED.

Claim boundary

Finite identities on an explicit family of packings, a pairing theorem for positive run sequences, and a named analytic remainder V_2 on a bordered-chain polynomial. No statement about zeros of $\zeta(s)$, in either direction. No RH claim.

References

- [1] S. Hamann, *A median cap for gaps of a tiled integer packing*, standalone note, August 2026, `rh/problem/medcap_lemma.tex`.